BACHY105 · APPLIED CHEMISTRY · VIT VELLORE · MODULE 4 (6 HOURS)

Spectroscopic and Diffraction Techniques

EM Spectrum & Space Payloads · UV–Visible Electronic Transitions, Spectral Shifts & Beer–Lambert Law · X-Ray Generation, Bragg's Law Derivation & Debye–Scherrer Crystallite Size · Complete Solved PYQs

Theme: **evs-vit.vercel.app Dark Minimalist** | Sections: **M4S1 – M4S6 (Zero Information Loss)** | Coverage: **Chandrayaan-3 & Mars Payloads, UV–Vis Shifts, Bragg Proof & Official FAT Questions**

M4S0 Quick-Reference Equations, Constants & Unit Conversions

Memory Strip

PLANCK–EINSTEIN PHOTON ENERGY

$E = h\nu = hc / \lambda = hc\,\tilde{\nu}$

$h = 6.626\times10^{-34}$ J s · $c = 3.0\times10^{8}$ m s^{-1} · E (eV) = 1240 / λ (nm)

BEER–LAMBERT LAW

$A = \log_{10}(I_0 / I) = -\log_{10} T = \varepsilon L c$

$T = I / I_0 = 10^{-A}$ | ε: L mol^{-1} cm^{-1} · L: cm · c: mol L^{-1}

BRAGG'S LAW OF X-RAY DIFFRACTION

$2 d_{hkl} \sin\theta = n\lambda$

Cubic crystal: $d_{hkl} = a / \sqrt{(h^2 + k^2 + l^2)}$ | Cu Kα: λ = 1.5406 Å (0.15406 nm)

DEBYE–SCHERRER CRYSTALLITE SIZE

$D = (k\lambda) / (\beta \cos\theta)$

k = 0.9 (0.89–0.94) | MUST convert FWHM β to radians: β(rad) = β(°) × (π / 180)

M4S1 Electromagnetic Radiation, 7-Region EM Spectrum & Space Exploration Spectrometers

7 EM Regions + Mars & Chandrayaan-3 Payloads

EM Spectral Region	Wavelength Range (λ)	Quantum Transition / Molecular Process	Analytical Technique
1. Gamma Rays (γ)	< 0.01 nm (< 0.1 Å)	Nuclear energy level transitions	Mössbauer spectroscopy (^{57}Fe mineral oxidation state)
2. X-Rays	0.01 – 10 nm (0.1 – 100 Å)	Inner-shell (K, L) core electron transitions & elastic scattering	**XRD** (crystal structure), XRF, XPS, APXS
3. Ultraviolet (UV)	**10 – 400 nm** (Quartz UV: 200–400 nm)	Valence electronic transitions (σ→σ*, n→σ*, π→π*, n→π*)	**UV–Visible absorption spectroscopy**
4. Visible Light	**400 – 800 nm** (Violet 400 → Red 800)	Conjugated π-electron & d–d charge-transfer transitions	Colorimetry, UV–Vis, LIBS atomic emission
5. Infrared (IR)	0.8 – 1000 μm (Mid-IR: 4000–400 cm^{-1})	Molecular vibrational & rotational transitions (Δμ ≠ 0)	FT-IR, Thermal Emission Spectroscopy (Mini-TES)
6. Microwaves	1 mm – 30 cm	Pure rotational transitions of polar molecules / electron spin flip	Rotational microwave & ESR/EPR spectroscopy
7. Radio Waves	> 30 cm (m)	Nuclear spin flips in an external magnetic field	NMR (^{1}H, ^{13}C) & MRI

6 Ways EM Radiation Interacts with Matter
Slide Concept

- (1) **Absorption** (photon energy promotes atom/molecule $E_1 \rightarrow E_2$; basis of UV–Vis & IR); (2) **Emission / Luminescence** (excited state relaxes $E_2 \rightarrow E_1$ emitting photon; basis of LIBS/AES/Fluorescence); (3) **Transmission & Refraction**; (4) **Reflection**; (5) **Elastic Scattering (Rayleigh & Thomson Scattering)** (no wavelength change; basis of **XRD**); (6) **Inelastic Scattering (Raman & Compton Scattering)**.

Atomic vs. Molecular Spectroscopy
Line vs. Broad Band Spectra

- **Atomic Spectroscopy (AAS, AES, LIBS, ICP-OES):** Probes isolated gas-phase atoms lacking vibrational or rotational sublevels. Electronic transitions between discrete atomic orbitals produce **sharp, narrow Line Spectra** characteristic of each element.
- **Molecular Spectroscopy (UV–Vis, IR):** Each electronic level (S_0, S_1) in a molecule contains a manifold of **vibrational (v) and rotational (J) sublevels**. Simultaneous electronic + vibrational + rotational transitions (*rovibronic transitions*) merge in solution to produce **broad Band Spectra** characterized by λ_{max} and ε_{max}.

Space Exploration Spectrometers
Mars Rovers & ISRO Chandrayaan-3

- **NASA Mars Exploration Rovers (Spirit, Opportunity, Curiosity, Perseverance):**
  • **Mini-TES (Miniature Thermal Emission Spectrometer):** Mid-IR spectrometer identifying carbonates, silicates, and aqueous clay minerals from a distance.
  • **Mössbauer Spectrometer (MB):** γ-ray spectrometer identifying iron-bearing minerals (**jarosite, hematite, goethite, olivine**) and Fe^{2+}/Fe^{3+} ratios — proving ancient acidic liquid water on Mars.
  • **APXS (Alpha Particle X-Ray Spectrometer):** Uses ^{244}Cm alpha particles/X-rays to induce characteristic X-ray fluorescence for elemental analysis of Martian rocks (Na to Br).
- **ISRO Chandrayaan-3 Lunar South Pole Mission (Vikram Lander & Pragyan Rover, Aug 2023):**
  • **LIBS (Laser-Induced Breakdown Spectroscope on Pragyan Rover):** Fires high-energy laser pulses to vaporize lunar regolith into plasma and records atomic emission lines — **first in-situ confirmation of Sulfur (S) near the lunar south pole** along with Al, Ca, Fe, Cr, Ti, Mn, Si, and O.
  • **APXS (on Pragyan Rover):** Elemental composition of lunar soil via X-ray fluorescence.
  • **Vikram Lander Payloads: ChaSTE** (lunar topsoil thermal profile), **ILSA** (lunar seismic activity), **RAMBHA-LP** (near-surface plasma density), and **SHAPE** (spectropolarimetry of Earth from lunar orbit).

M4S2 UV–Visible Spectroscopy: 4 Electronic Transitions, Spectral Shifts & Color Wheel

Energy Order + 4 Spectral Shifts + Aniline & Pyridine Cases

4 Electronic Transitions & Strict Energy Order
σ → σ* > n → σ* > π → π* > n → π*

- **Strict Energy Ranking: $\Delta E(\sigma \rightarrow \sigma^*) > \Delta E(n \rightarrow \sigma^*) \approx \Delta E(\pi \rightarrow \pi^*) > \Delta E(n \rightarrow \pi^*)$** (consequently λ_{max} follows the exact reverse order).
- **1. σ → σ* (Highest ΔE, Vacuum UV < 150 nm):** Excitation of a bonding σ electron in saturated alkanes (CH_4 at 125 nm, C_2H_6 at 135 nm). Transparent in standard 200–800 nm UV–Vis, making **hexane and cyclohexane ideal UV solvents**.
- **2. n → σ* (150 – 250 nm, ε = 100 – 3,000 L mol^{-1} cm^{-1}):** Saturated compounds containing heteroatoms with non-bonding lone pairs (**–OH, –NH_2, –SH, –Cl, –Br, –I**), e.g., CH_3OH (183 nm), CH_3Cl (173 nm), CH_3I (258 nm).
- **3. π → π* (200 – 700 nm, Symmetry Allowed, High ε = 10,000 – 50,000):** Unsaturated molecules with double/triple bonds or aromatic rings (**C=C, C≡C, C=O, benzene**). Ethylene absorbs at 171 nm; extended conjugation lowers ΔE into the near-UV/visible (1,3-butadiene: 217 nm; β-carotene with 11 conjugated double bonds: 455 nm, orange).
- **4. n → π* (270 – 350 nm, Lowest ΔE, Symmetry Forbidden, Low ε = 10 – 100):** Promotion of a lone-pair electron on O/N/S to an antibonding π* orbital in unsaturated heteroatom groups (**C=O in acetone at 280 nm, ε ≈ 15; –N=O; –N=N–**).

Selection Rules, Chromophores & Auxochromes
Why π→π* is Intense & n→π* is Weak

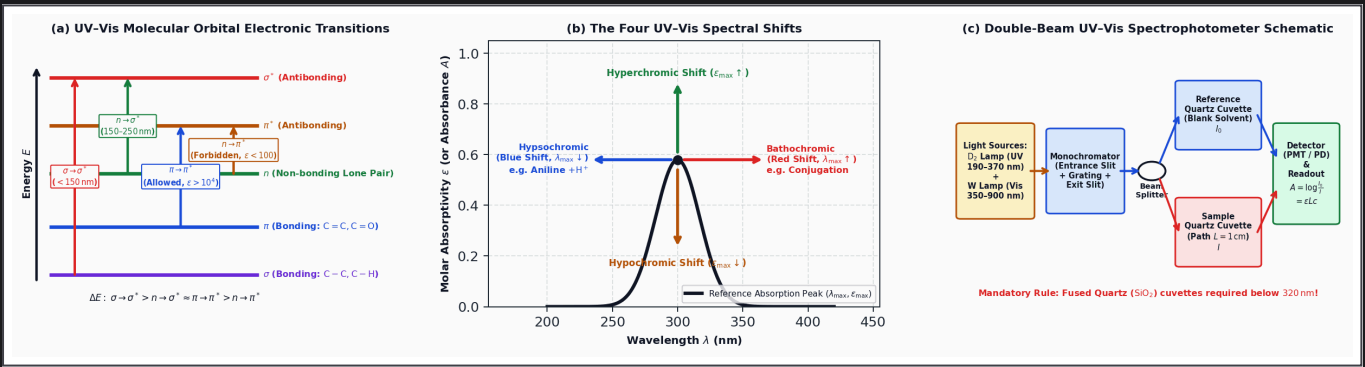
- **Selection Rules:** (1) **Spin Rule (ΔS = 0):** Singlet → Singlet ($S_0 \rightarrow S_1$) is spin-allowed; Singlet → Triplet ($S_0 \rightarrow T_1$) is spin-forbidden; (2) **Laporte / Spatial Overlap Rule:** In **π → π***, both orbitals lie in the same plane with strong spatial overlap (**symmetry-allowed, ε > 10^4**); in **n → π***, the non-bonding p-orbital is orthogonal (90°) to the π* orbital, giving near-zero transition dipole moment (**symmetry-forbidden, ε = 10–100**).
- **Chromophore ("Color-Bearer"):** Covalently unsaturated functional group responsible for electronic absorption in the UV–Vis region (e.g., `C=C, C≡C, C=O, –NO₂, –N=N–, quinonoid ring`).
- **Auxochrome ("Color-Enhancer"):** Saturated substituent with lone pairs (**–OH, –OR, –NH_2, –NR_2, –SH, –X**) that does not absorb above 200 nm by itself, but when attached to a chromophore extends conjugation via resonance (+M effect), producing a **bathochromic (red) and hyperchromic shift** (e.g., Benzene λ_{max} = 254 nm → Phenol 270 nm → Aniline 280 nm).

The 4 Spectral Shifts & Official Slide Case Studies
Aniline Acid Shift & 2-Methylpyridine

- **1. Bathochromic Shift (Red Shift):** Shift of λ_{max} toward **longer wavelength** (lower energy) due to extended conjugation, auxochrome attachment, or deprotonation of phenol (Phenol 270 nm → Phenoxide ion $C_6H_5O^-$ 287 nm in NaOH).
- **2. Hypsochromic Shift (Blue Shift):** Shift of λ_{max} toward **shorter wavelength** (higher energy) due to loss of conjugation, protonation of lone pair, or polar H-bonding stabilizing the n-orbital.
  • **Official Slide Case 1 — Aniline in Acidic Solution (280 nm → 265 nm Blue Shift):** In neutral medium, the lone pair on the $-NH_2$ group of **Aniline ($C_6H_5NH_2$)** conjugates with the benzene π-ring (+M effect), giving **λ_{max} = 280 nm (ε = 1430)**. In acidic medium (HCl), protonation forms the **Anilinium ion ($C_6H_5NH_3^+$)**, binding the lone pair to H^+ and destroying n–π conjugation; the spectrum reverts to benzene-like absorption at **λ_{max} = 254–265 nm (ε = 160)** (both a **hypsochromic and hypochromic shift**!).
- **3. Hyperchromic Shift (↑ ε_{max}) & 4. Hypochromic Shift (↓ ε_{max}):** Increase vs. decrease in molar absorptivity intensity ε_{max}.
  • **Official Slide Case 2 — Pyridine vs. 2-Methylpyridine:** Introducing an electron-donating methyl group in **2-methylpyridine** increases transition probability and produces a **hyperchromic shift** (ε_{max} rises from 2750 in pyridine to 3560 L mol^{-1} cm^{-1}) with a slight bathochromic shift.

Complementary Color Wheel & Visible Indicators
Absorbed vs. Perceived Color

- A compound appears as the **complementary color** of the wavelength it absorbs:
  • Absorbs **400–435 nm (Violet)** → Appears **Yellow-Green** | • Absorbs **435–480 nm (Blue)** → Appears **Yellow** (e.g., **β-Carotene / Tartrazine** at ~450 nm)
  • Absorbs **490–500 nm (Blue-Green)** → Appears **Red** (e.g., **Congo Red** at ~497 nm)
  • Absorbs **500–560 nm (Green)** → Appears **Purple / Magenta** (e.g., $KMnO_4$ at 525 nm)
  • Absorbs **580–595 nm (Yellow)** → Appears **Blue** | • Absorbs **600–680 nm (Orange-Red)** → Appears **Blue / Blue-Green** (e.g., **Methylene Blue** absorbs at **664 nm** → appears intense **Blue**; **Chlorophyll a** absorbs Blue ~430 nm & Red ~662 nm and reflects **500–560 nm → appears Green**).



Visual Summary M4.1 (Slide Diagrams): (Left) Molecular orbital energy ladder showing σ→σ*, n→σ*, π→π*, and n→π* transitions; (Center) The four UV–Vis spectral shifts (Bathochromic red, Hypsochromic blue, Hyperchromic ↑ε, Hypochromic ↓ε); (Right) Double-beam UV–Vis spectrophotometer optical path with D_2/W lamps, monochromator, rotating chopper, reference/sample quartz cuvettes, and photodetector.

M4S3 Beer–Lambert Law, Double-Beam Instrumentation & Analytical Applications

Law + Cuvette Rule + Keto–Enol Tautomerism

Beer–Lambert Law & Deviations
A = ε L c

- **Combined Statement:** When a monochromatic beam of intensity I_0 passes through a homogeneous absorbing solution of concentration c (mol L^{-1}) and path length L (cm), the absorbance is directly proportional to both L (Lambert's law) and c (Beer's law): **$A = \log_{10}(I_0 / I) = -\log_{10} T = \varepsilon L c$**.
- **Conditions & Limitations (Deviations from Linearity):** Valid only for **dilute solutions (c ≤ 0.01 M)** and **monochromatic radiation**. Deviations occur at high concentrations due to solute dimerization/association (e.g., methylene blue dimers), refractive index changes, chemical equilibria ($2CrO_4^{2-} + 2H^+ \rightleftharpoons Cr_2O_7^{2-} + H_2O$), fluorescence, or stray polychromatic light.

UV–Vis Spectrophotometer Components
Why Quartz Cuvettes Below 320 nm

- **1. Radiation Sources: Deuterium (D_2) or Hydrogen discharge lamp** for the UV region (**190 – 370 nm**) + **Tungsten (W) filament / Tungsten-halogen lamp** for the Visible/Near-IR region (**350 – 900 nm**).
- **2. Monochromator:** Entrance slit + reflection diffraction grating (or quartz prism) + exit slit to isolate a narrow wavelength band ($\Delta\lambda \approx 1$ nm).
- **3. Sample & Reference Cuvettes (1.0 cm path): Mandatory Exam Rule: Fused Quartz (Silica) cuvettes** MUST be used in the UV region (< 320 nm) because ordinary silicate glass and plastic strongly absorb UV radiation below 320 nm.
- **4. Double-Beam vs. Single-Beam Advantage:** A rotating sector mirror (chopper) splits the monochromatic beam continuously between the **Reference cuvette (pure solvent blank, I_0)** and **Sample cuvette (analyte solution, I)**, automatically canceling lamp drift, solvent absorption, and cuvette reflection losses in real time.

5 Key Analytical Applications of UV–Vis
Keto–Enol & Cis–Trans

- **1. Quantitative Trace Analysis:** Calibration curve of A vs. c (slope = εL) determines unknown concentrations of drugs, metal complexes, and pollutants down to 10^{-6} M.
- **2. Keto–Enol Tautomerism in Ethyl Acetoacetate (Official Slide Example):** The **Keto form** (unconjugated C=O) shows only a weak n → π* band at **λ_{max} = 275 nm (ε ≈ 16)**, whereas the **Enol form** (intramolecularly H-bonded conjugated `HO–C=CH–C=O`) exhibits an intense symmetry-allowed π → π* band at **λ_{max} = 245 nm (ε ≈ 18,000)** in non-polar hexane!
- **3. Geometrical Isomerism (Cis vs. Trans Stilbene): trans-Stilbene** is planar with full π-overlap (**λ_{max} = 295 nm, ε = 27,000**), whereas **cis-Stilbene** suffers steric crowding between ortho-hydrogens that twists the phenyl rings out of plane, reducing conjugation (**λ_{max} = 280 nm, ε = 10,500**).
- **4. Distinction of Vitamin A_1 vs. Vitamin A_2:** Vitamin A_1 (5 conjugated C=C bonds) absorbs at **325 nm**, whereas Vitamin A_2 (6 conjugated C=C bonds) absorbs at **351 nm**.
- **5. Chemical Kinetics & Charge-Transfer (CT) Complexes:** Real-time monitoring of dA/dt and detection of I_2–benzene or metal–ligand CT bands.

M4S4 X-Ray Generation, Crystalline vs. Amorphous Solids & Bragg's Law Derivation

Cu Kα Tube + Step-by-Step Geometric Proof

X-Ray Tube Generation & Cu Kα Monochromation
λ = 1.5406 Å (8.04 keV)

- **Evacuated X-Ray Tube:** Electrons emitted thermionically from a heated **Tungsten (W) filament cathode** are accelerated across a high DC potential (**30–45 kV**) in high vacuum to bombard a water-cooled **Copper (Cu) or Molybdenum (Mo) target anode**; X-rays exit through low-Z **Beryllium (Be) windows**.
- **1. Continuous Spectrum (Bremsstrahlung / Braking Radiation):** High-speed electrons decelerating in the nuclear field of Cu emit a continuous spectrum with a sharp short-wavelength cutoff **λ_{min} (Å) = hc / (eV) = 12,400 / V(volts)**.
- **2. Characteristic Line Spectrum (Kα and Kβ):** When an incident electron knocks out a core **K-shell (n=1)** electron of Cu, an outer electron drops down to fill the vacancy:
  • **L-shell (n=2) → K-shell (n=1):** Emits intense **Cu Kα radiation (λ = 1.5406 Å, E = 8.04 keV)**.
  • **M-shell (n=3) → K-shell (n=1):** Emits higher-energy **Cu Kβ radiation (λ = 1.392 Å)**.
  • A thin **Nickel (Ni) β-filter foil** (whose K-absorption edge lies between Cu Kα and Kβ) absorbs Kβ and leaves monochromatic **Cu Kα** for XRD.

Crystalline vs. Amorphous SiO_2
Quartz vs. Silica Glass

- **Crystalline Solids (α-Quartz & Cristobalite SiO_2):** Possess periodic **long-range 3D atomic order** over thousands of unit cells, anisotropic properties, sharp melting point, and produce **sharp, discrete Bragg peaks** in XRD at specific 2θ angles.
- **Amorphous Solids (Fused Silica Glass SiO_2):** Possess only short-range tetrahedral SiO_4 order without periodic long-range planes, isotropic properties, gradual glass-transition softening, and produce only a **single broad diffuse halo around 2θ ≈ 21–22°**.

Step-by-Step Geometric Derivation of Bragg's Law
$2\,d_{hkl}\sin\theta = n\lambda$ · 5M Proof

- **Step 1 (Setup):** Consider a parallel monochromatic X-ray beam of wavelength **λ** incident at a glancing angle **θ** (Bragg angle, measured from the crystal plane) on a family of parallel Miller lattice planes **(hkl)** separated by interplanar spacing $\mathbf{d_{hkl}}$.
- **Step 2 (Path Difference):** Ray 1 scatters specularly off atom A in the top plane; Ray 2 penetrates to the second plane and scatters off atom B directly below A. Drop perpendiculars AC and AD from A onto the incident and reflected paths of Ray 2.
- **Step 3 (Trigonometry in Right Triangles ΔACB and ΔADB):** Since AB = d_{hkl} and ∠CAB = ∠DAB = θ, the extra path lengths traveled by Ray 2 are **CB = d sin θ** and **BD = d sin θ**.
- **Step 4 (Total Path Difference Δx): $\Delta x = CB + BD = 2\,d_{hkl}\sin\theta$**.
- **Step 5 (Constructive Interference Condition):** Scattered waves remain in phase and produce an intense diffraction peak only when the total path difference equals an integer multiple of the wavelength: **$2\,d_{hkl}\sin\theta = n\lambda$** (where n = 1, 2, 3... is the diffraction order). Note: Since sin θ ≤ 1, diffraction requires **$\lambda \leq 2d$** (~1–2 Å X-rays match atomic spacings).

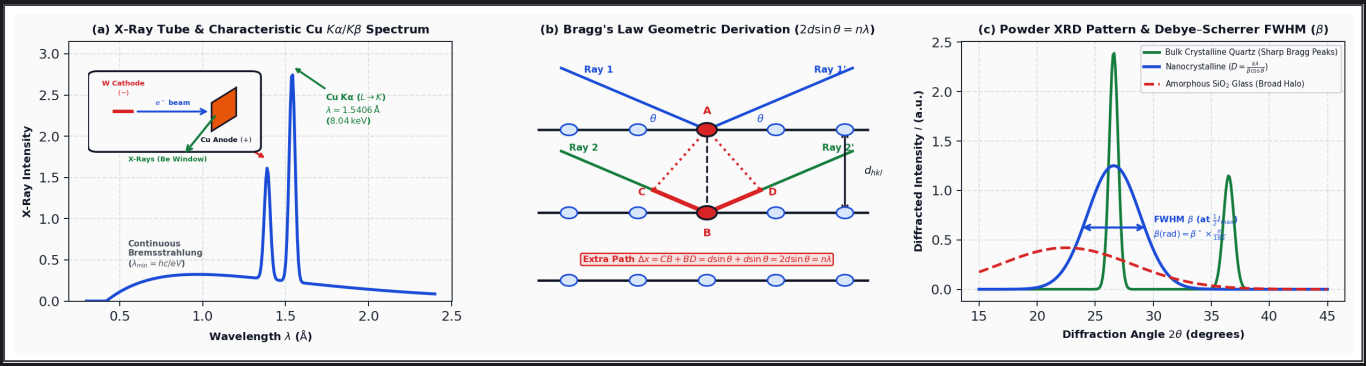
M4S5 Powder Diffractometer, Microstructural Hierarchy & Debye–Scherrer Equation

Unit Cell vs. Crystallite vs. Grain vs. Particle

Structural Entity	Typical Length Scale	Physical Definition	Primary Characterization Tool
1. Unit Cell	**0.3 – 2.0 nm (3 – 20 Å)**	Smallest repeating 3D parallelepiped (a, b, c, α, β, γ) building block of a crystal	**XRD** peak positions (2θ → d_{hkl} → lattice constants)
2. Crystallite (Coherent Domain)	**2 – 100 nm**	Single coherently diffracting crystal domain with identical lattice orientation	**XRD Peak Broadening (Debye–Scherrer Equation)**
3. Grain	**100 nm – 100 µm**	Individual crystalline region bounded by grain boundaries (may contain 1 or more crystallites)	**SEM / EBSD** & Optical Metallography
4. Particle / Agglomerate	**10 nm – mm**	Physical powder unit composed of multiple fused grains/crystallites	**TEM, SEM & Dynamic Light Scattering (DLS)**

Powder XRD Goniometer & Debye–Scherrer Law
$D = k\lambda / (\beta\cos\theta)$

- **θ–2θ Bragg–Brentano Goniometer:** Finely ground powder contains millions of randomly oriented crystallites so every (hkl) plane satisfies Bragg's condition for some fraction of grains. As the sample rotates by angle **θ** (or X-ray source by θ), the detector rotates by **2θ**, plotting Intensity vs. **2θ**. Phase identification matches d-spacings and relative intensities (I/I_0) against the **JCPDS / ICDD Powder Diffraction File (PDF) database**.
- **Debye–Scherrer Equation for Nanocrystallite Size (D < 100 nm):** Finite crystal domains lack enough parallel planes to cancel slightly off-Bragg rays, causing inverse peak broadening: **$D = (k\lambda) / (\beta\cos\theta)$**, where **k ≈ 0.9** is the dimensionless shape factor, **λ** is X-ray wavelength, **θ** is the Bragg angle (**half of the 2θ peak position!**), and **β** is the corrected Full Width at Half Maximum (FWHM) in **radians**: **$\beta(rad) = [\sqrt{(\beta_{measured}^2 - \beta_{instrument}^2)}] \times (\pi / 180)$**.
- **Strengths & Limitations of XRD:** *Pros:* Non-destructive, rapid phase & polymorph ID (rutile vs. anatase TiO_2), lattice constants to ±0.0001 Å, and average nanocrystallite size. *Cons:* Insensitive to amorphous phases at low % (<2–5 wt%), weak scattering from light elements (H, Li), and peak overlap in low-symmetry multiphase mixtures.



Visual Summary M4.2 (Slide Diagrams): (Left) Geometric ray diagram of Bragg's Law showing path difference CB + BD = 2d sin θ = nλ between parallel (hkl) planes; (Center) Sharp crystalline quartz Bragg peaks vs. broad amorphous silica glass halo; (Right) Debye–Scherrer peak broadening showing how smaller nanocrystallites (D) produce wider FWHM (β).

M4S6 Complete Solved Numericals & Official VIT FAT Question Bank Blueprints

4 Numericals + Official Q1–Q5

Solved Numerical 4.1 — Beer–Lambert Law: Absorbance, % Transmittance & Unknown Concentration

CAT-2 / FAT Numerical · 5 Marks

A 5.0 × 10^{-5} M solution of an organic dye in a 1.0 cm quartz cuvette exhibits 25.0% transmittance at λ_{max} = 520 nm. Calculate (a) absorbance A, (b) molar absorptivity ε, and (c) concentration of an unknown sample that gives 60.0% transmittance in the same cell.

(a) Absorbance (A_1): T_1 = 0.250 ⇒ **$A_1 = -\log_{10}(0.250) = 0.6021$**.
(b) Molar Absorptivity (ε): **$\varepsilon = A_1 / (L\, c_1) = 0.6021 / (1.0\ cm \times 5.0 \times 10^{-5}\ mol\ L^{-1}) = 1.204 \times 10^4\ L\ mol^{-1}\ cm^{-1}$**.
(c) Unknown Concentration (c_2 at T_2 = 0.600): $A_2 = -\log_{10}(0.600) = 0.2218$ ⇒ **$c_2 = A_2 / (\varepsilon L) = 0.2218 / (1.204 \times 10^4 \times 1.0) = 1.84 \times 10^{-5}$ M**.

Solved Numerical 4.2 — Bragg's Law: Interplanar Spacing d_{hkl} & Cubic Lattice Parameter (a)

Official Question Bank · 5 Marks

*First-order (n = 1) X-ray diffraction from the (111) plane of a face-centered cubic (FCC) copper crystal is observed at a diffraction angle **2θ = 43.30°** using Cu Kα radiation (λ = 1.5406 Å). Calculate (a) interplanar spacing d_{111}, (b) cubic unit-cell edge length a, and (c) atomic radius of Cu.*

1. Bragg Angle θ: Given 2θ = 43.30° ⇒ **θ = 21.65°** ⇒ sin(21.65°) = 0.3689.
(a) Interplanar Spacing (d_{111}): **$d_{111} = n\lambda / (2 \sin\theta) = (1 \times 1.5406\ Å) / (2 \times 0.3689) = 2.088\ Å\ (0.2088\ nm)$**.
(b) Cubic Lattice Constant (a): For (111), $\sqrt{(h^2 + k^2 + l^2)} = \sqrt{(1+1+1)} = \sqrt{3} = 1.7321$ ⇒ **$a = d_{111}\sqrt{3} = 2.088 \times 1.7321 = 3.617$ Å**.
(c) FCC Atomic Radius: r = a / (2√2) = 3.617 / 2.8284 = **1.279 Å**.

Solved Numerical 4.3 — Debye–Scherrer Calculation of Nanocrystallite Size (D)

High-Frequency FAT Numerical · 5 Marks

*An XRD peak of TiO_2 nanoparticles recorded with Cu Kα radiation (λ = 0.15406 nm) appears at **2θ = 25.30°** with a Full Width at Half Maximum (FWHM) of **β = 0.45°**. Taking shape factor k = 0.90, calculate the average crystallite size D in nm.*

Step 1 (Halve 2θ to get θ): θ = 25.30° / 2 = **12.65°** ⇒ cos(12.65°) = **0.9757**.
Step 2 (Convert FWHM β from degrees to radians): **$\beta(rad) = 0.45° \times (\pi / 180°) = 0.45 \times 0.017453 = 7.854 \times 10^{-3}$ rad**.
Step 3 (Debye–Scherrer Formula): **$D = (k\lambda) / (\beta \cos\theta) = (0.90 \times 0.15406\ nm) / (7.854 \times 10^{-3} \times 0.9757) = 0.13865 / 0.007663 = 18.09$ nm**.